

Multi-Agent System for **Data Quality**

Automated detection and remediation of dirty datasets for
NoiPA

TEAM

Eliya Allam · Simone Filosofi · Mattia Cervelli

COURSE

Machine Learning · LUISS × Reply

YEAR

2026

Approach & design philosophy



Semi-RAG semantic resolution

`embed → retrieve top-5 → LLM re-ranks`



Deterministic safety nets

`code first, LLM only when needed`



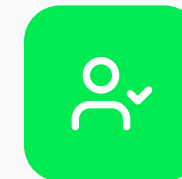
Prompt stratification

`one system prompt per agent / tool`



Baseline as ground truth

`noipa_schema_registry.json`



Human-in-the-loop

`propose → sandbox → review → approve`

Two custom knowledge bases

FILE · 01

noipa_schema_registry.json

4 **15** **143** **12**
DOMAINS DATASETS COLUMNS \$REF DEFS

- Format specs · enum / regex / range / date
- Domain observations & conventions

→ used by: all agents

FILE · 02

column_descriptions.json
+ embeddings.pkl

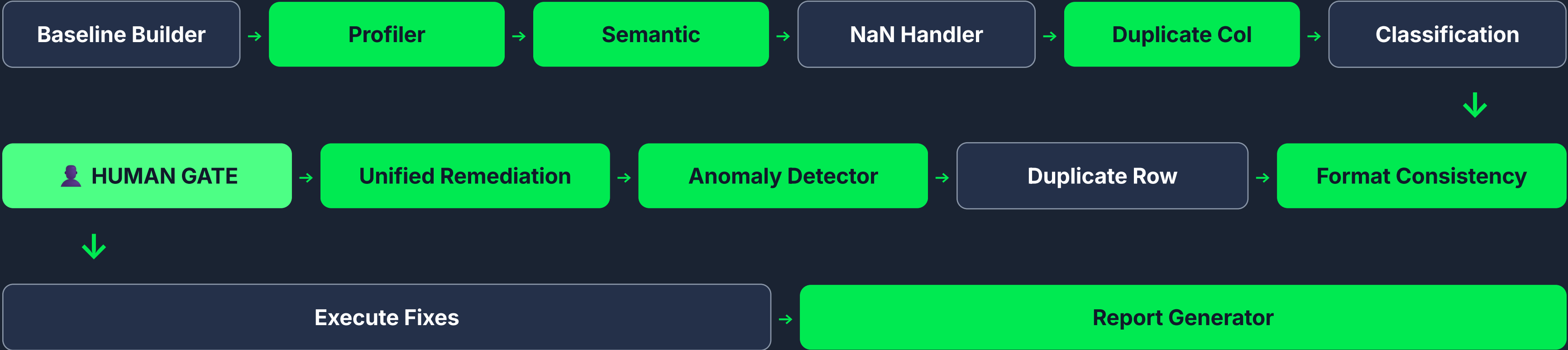
54 **1536**
CANONICAL COLS EMBED DIM

- Descriptions + sample values per column
- Loaded once at startup · cached to disk

→ used by: Semantic Agent (retrieval)

“Building these two files was the highest-leverage investment in the system.”

Pipeline architecture



☒ LLM call ☐ Deterministic ☒ Human gate Shared state: PipelineState (Pydantic) · Orchestration: [LangGraph](#)

Baseline Builder

NO LLM

```
# step 1
noipa_schema_registry.json → resolve $ref → validate → BaselineFile
↓
baseline.json + state.baseline
```

- Resolves \$ref across 12 shared definitions
- Validates into typed Pydantic BaselineFile
- All downstream agents query this compiled object

Profiler

LLM • 1 CALL

```
columns + samples + baseline signatures
```

```
↓ LLM
```

```
detected_domain + detected_language
```

- Builds **domain signature map** → LLM with structured output
- Gates downstream validation — specs differ by domain

Semantic Resolver

LLM · N+1 CALLS

①

Batch describe

all columns → LLM
one-sentence desc each

②

Embed & retrieve

text-embedding-3-small
cosine vs 54 baseline
top-5 + sample/dtype
boost

③

LLM verdict

top-5 + 30-row sample
→ canonical_match
dtype · placeholders
related · casing

EMBEDDINGS

④

Validate & cast

canonical validated
dtype cast in-place

output → state.payload - ColumnPayload per column

NaN Handler

NO LLM

```
payload.placeholders → replace in dataset → pd.NA  
↓  
baseline is_nullable=False? → ValidationReport violation
```

- Placeholders come from Semantic Agent — **no extra LLM call**
- Catches disguised NaNs: "N/A", "-", "NULL", "missing"

Duplicate Column Resolver

LLM • 1 CALL / GROUP

```
all columns → Jaccard value-overlap (threshold 0.80)
↓
duplicate groups
↓
survivor = fewest nulls → backfill from siblings
↓
LLM elects canonical name → drop siblings → rename
```

- Same column under different labels — e.g. cod_ente vs ente_code
- LLM used for **name election only**, not merging logic

Classification

STUB

```
surviving columns → lowercase names → state.classifications
```

→ LLM call planned — snake_case normalization per registry convention. Not yet active.

Format Consistency

LLM · 2 CALLS / COLUMN

LEFT · VALIDATION & CORRECTION

sample values → profile format (deterministic)

↓

LLM refines spec (infer_format_spec)

↓

validate values → FormatViolation list

↓

LLM proposes corrections

RIGHT · IMPUTATION HINTS

deterministic

columns with NaNs

→ test related_columns as predictors

→ compute purity + coverage

→ emit ImputationHint

outputs → validation_reports · value_corrections · inferred_format_specs · imputation_hints

Duplicate Row Removal

NO LLM

```
dataset (post-normalization) → df.drop_duplicates() → deduplicated dataset
```

→ Runs last so comparisons are on clean, normalized values

Anomaly Detector

LLM • COMMENTARY ONLY

DETECTION • DETERMINISTIC

- **Numeric** — IQR outlier detection
- **Categorical** — rare-category frequency
- **Categorical** — most-frequent-value analysis
- identifier columns excluded

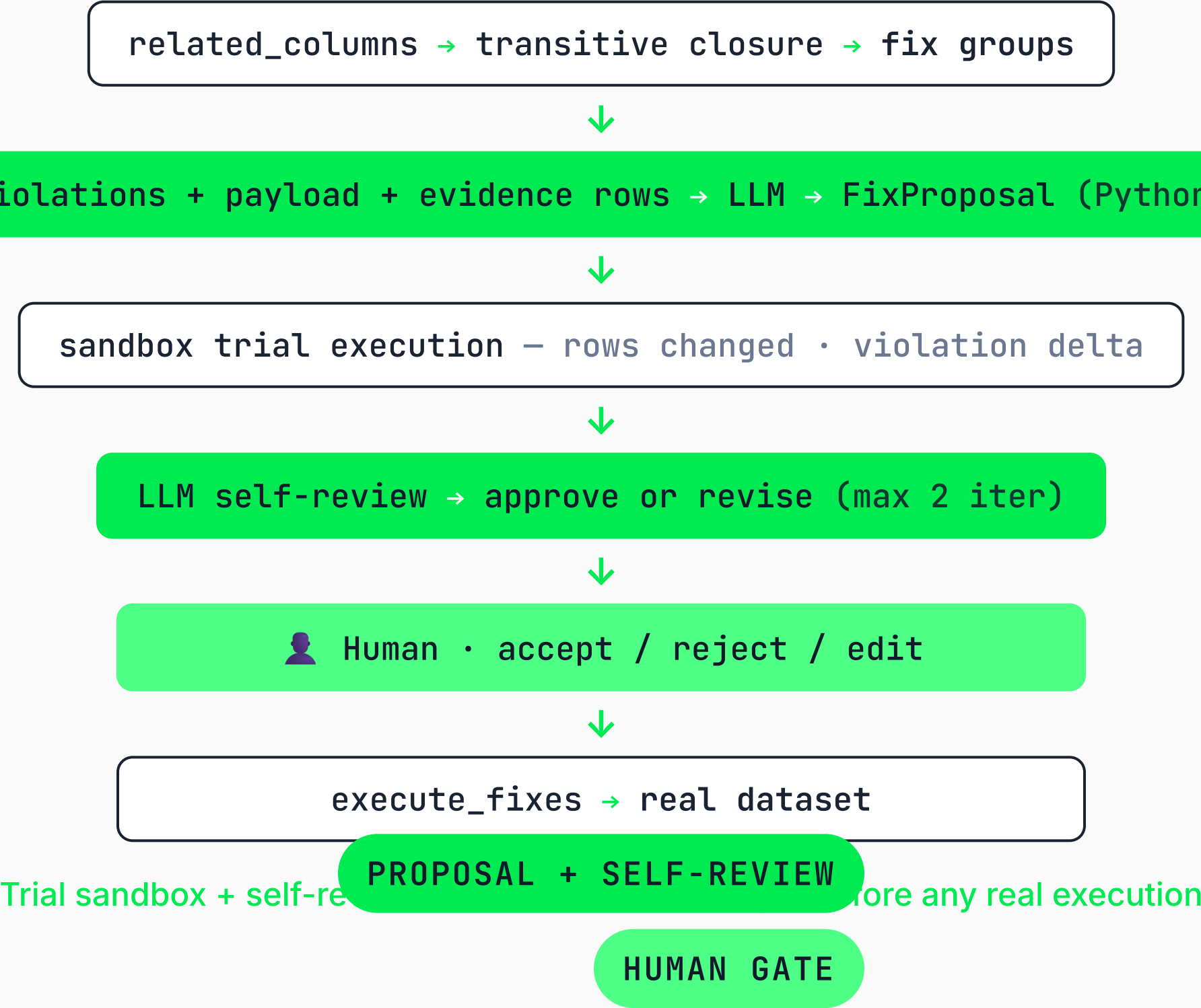
LLM ROLE

anomaly group → LLM → narrative comment only

no automated fix • **human decides**

output → state.anomaly_reports — AnomalyReport + AnomalyEntry

Unified Remediation



Strengths, weaknesses, roadmap

✓ Strengths

Schema registry as ground truth · no LLM invention

Semi-RAG matching beats name-only on synonyms / IT-EN

Modular LangGraph DAG · auditable, testable

Deterministic-first, LLM only where semantics needed

Cached 54×1536 index · zero per-run retrieval cost

End-to-end · raw CSV → cleaned data + PDF report

⚠ Weaknesses

Evaluation scope · manual ground truth caps reproducibility

Sequential execution · independent agents could run in parallel

KB coverage · 54 hand-curated entries, gaps fall back to NaN

LLM coupling · quality tied to gpt-4o-mini, no offline replay yet

➔ Roadmap

Synthetic error injection · scalable recall eval

Learn anomaly thresholds from labelled history

Parallelise independent agents via LangGraph

Auto-expand KB · mine new datasets vs registry

Smarter imputation tier · confidence-gated LLM

Feedback loop · human reviews → KB